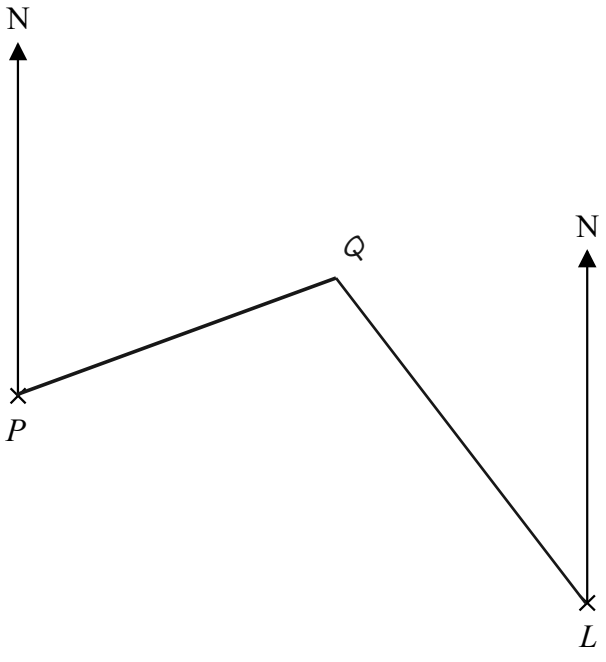
8 The accurate scale drawing shows the positions of port P and a lighthouse L.



Scale: 1 cm represents 4 km.

Aleena sails her boat from port P on a bearing of 070°

She sails for $1\frac{1}{2}$ hours at an average speed of 12 km/h to a port Q.

Find

(i) the distance, in km, of port Q from lighthouse L,

(ii) the bearing of port Q from lighthouse L.

PQ = 12 × 1½ = 18 km.

On the map, PQ = 18 ÷ 4 = 4.5 cm.

At 070° from P, mark Q 4.5 cm away.

Measure QL ≈ 5.4 cm → about 22 km.

Bearing of Q from L ≈ 322°.

5 marks total
Draw 070° from P
Calculate 18 km, then 4.5 cm
Locate Q on full diagram
QL ≈ 22 km
Bearing ≈ 322°

distance QL = ... km

bearing of Q from L = ... °

(Total for Question 8 is 5 marks)